Smart Audio Diagnostics - Analysis Report

21 Oct 2026, 21:52

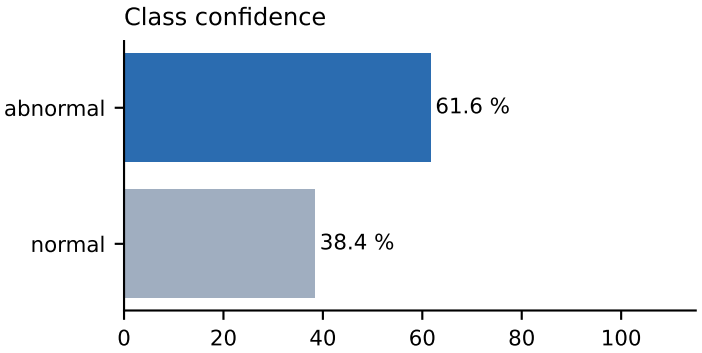
Group 09 · Digital Signal Processing · ECE, MNNIT Allahabad

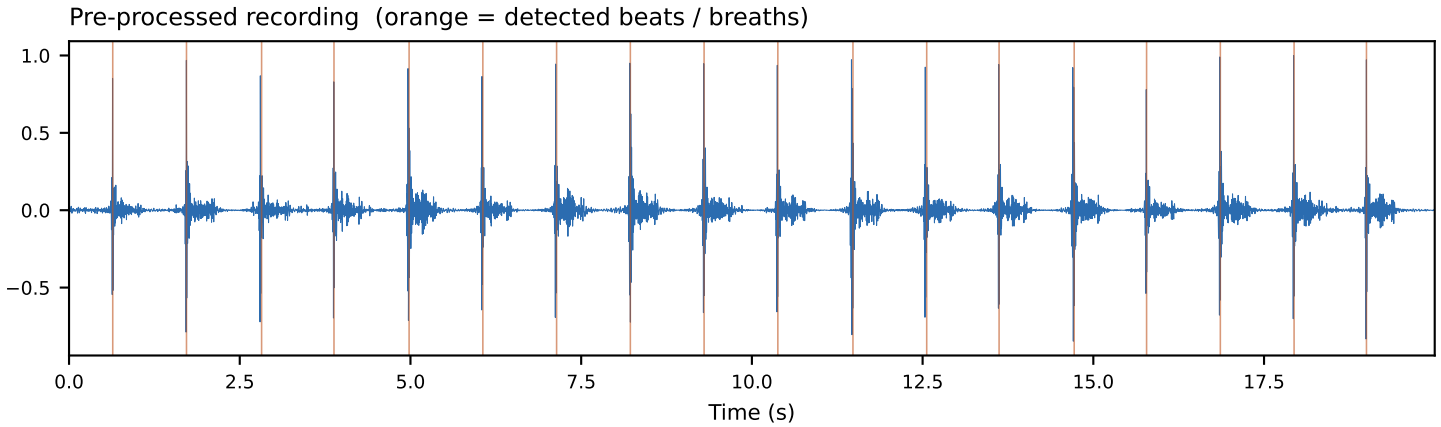
File	heart_abnormal_01.wav
Signal type	Medical - heart sound
Auto-detected as	heart (99 % confidence)
Classifier	Random Forest (held-out accuracy 85.7 %)
Trained on	real:C:\dsp\data\physionet2016

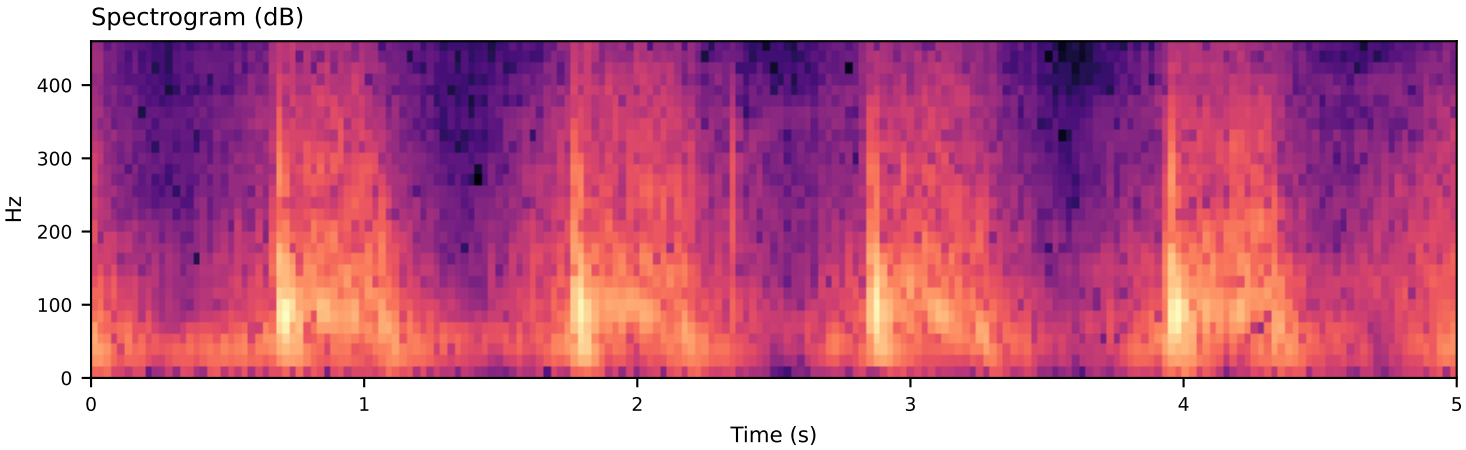
RESULT

ABNORMAL

Heart rate
56 bpm







DSP feature	Value	DSP feature	Value
Sampling frequency	2000 Hz	Spectral roll-off 85%	93.2 Hz
Band-pass	25-400 Hz	Spectral flatness	0.0023
Frame / hop	50 / 25 ms	Zero-crossing rate	0.0714
Mel filters	26	Short-time energy	0.00659
Spectral centroid	67.4 Hz	RMS	0.0356
Spectral bandwidth	34.1 Hz	MFCC-1 (mean)	-83.340

Screening / signal-classification output only. This is NOT a clinical diagnosis. Refer to a qualified clinician.

Generated by Smart Audio Diagnostics (Python · NumPy · SciPy · scikit-learn). MFCC computed from first principles.